Multi-Document Query & Common Section Summarization Agent

1. Overview

You are required to design and implement a **Multi-Document Intelligence Agent** that:

- Accepts multiple uploaded documents (PDF, DOCX, TXT, etc.)
- Answers user queries based on the content of those documents
- Identifies and summarizes common sections across documents
- Handles cases where the requested topic may or may not exist in the documents

The objective is to evaluate your ability to design document processing systems, implement retrieval-based pipelines, and build reliable summarization workflows.

2. Problem Statement

Build a system where:

1. A user uploads multiple documents.
2. The system processes and indexes them.
3. The user asks a question (e.g., "What are the security requirements?").
4. The system:
    a. Searches across all documents.
    b. Extracts relevant sections.
    c. Generates a consolidated answer.
    d. Provides a summary of the common themes across documents related to the topic.
5. If the topic is not present in any document, the system must clearly state that.

3. Scope of Work

A. Document Upload & Processing

- Accept multiple documents (minimum: PDF support required).
- Extract text content.
- Handle:
    - Structured text
    - Headings and sections
    - Multi-page documents
- Preprocess and index content for search.

B. Query Processing

- Accept a natural language query.
- Search across all uploaded documents.
- Retrieve relevant sections.
- Generate:
    - A direct answer to the query.
    - A summary of relevant content.
    - A "common insights" summary if multiple documents discuss the topic with reference.

C. Common Section Identification

The system should:

- Identify overlapping themes across documents.
- Detect similar sections (e.g., "Security Policy" across multiple PDFs).
- Produce:
    - A combined summary
    - A comparison view (optional but preferred)

D. Missing Topic Handling

If the queried topic does not exist in any document:

- The system must explicitly state:

"The topic was not found in the uploaded documents."

- Avoid hallucinated or fabricated content.

4. Example Use Case Flow

Scenario 1 – Topic Present in Multiple Documents

1. User uploads 3 policy documents.
2. User asks:

"Summarize data security requirements."

3. System:
    a. Finds relevant sections in all documents.

b. Combines insights.
c. Returns:
    i. A consolidated answer.
    ii. A summarized comparison of key points.

Scenario 2 – Topic Present in One Document Only

1. User uploads 5 documents.
2. User asks:

   "What is the disaster recovery strategy?"

3. System:
    a. Finds content in only 1 document.
    b. Responds with:
        i. Extracted answer.
        ii. Summary.
        iii. Note that other documents do not mention it.

Scenario 3 – Topic Not Found

1. User uploads documents.
2. User asks:

   "Explain blockchain integration."

3. System responds:
   "No references to blockchain integration were found in the uploaded documents."

5. Functional Requirements

Document Handling

- Support multiple document uploads.
- Extract text reliably.
- Handle at least 50 pages cumulatively.

Indexing & Retrieval

- Implement a search mechanism (keyword-based, semantic, or hybrid).
- Efficient retrieval of relevant chunks.

Summarization

- Generate:
  - Query-based summary
  - Cross-document common summary
- Avoid repetition.
- Keep summaries concise and structured.

Answer Generation

- Provide clear, structured responses.
- Reference document sources (preferred).

Error Handling

- Corrupted file handling.
- Unsupported format handling.
- Empty document handling.

6. Non-Functional Requirements

- Response time under 5 seconds (for moderate dataset).
- Scalable to large documents.
- No hallucinated answers.
- Clear traceability to document source.
- Secure document storage.

7. Deliverables

Submit:

1. Source Code (Git repository)
2. README including:
   a. Setup instructions
   b. Architecture diagram
   c. Design decisions
   d. Assumptions
3. Example test documents (small sample set)